

Gradient Descent :- (Polynomial Regression).

$$y_i = ax_i^2 + bx_i + c$$

$$E = \frac{1}{n} \sum_{i=0}^n (y_i - (ax_i^2 + bx_i + c))^2$$

$$\frac{dE}{da} = -\frac{2}{n} \sum_{i=0}^n x_i^2 (y_i - (ax_i^2 + bx_i + c))$$

$$\frac{dE}{db} = -\frac{2}{n} \sum_{i=0}^n x_i (y_i - (ax_i^2 + bx_i + c))$$

$$\frac{dE}{dc} = -\frac{2}{n} \sum_{i=0}^n (y_i - (ax_i^2 + bx_i + c))$$

$$\frac{dE}{da} = g_a, \quad \frac{dE}{db} = g_b, \quad \frac{dE}{dc} = g_c$$

$$\therefore a_{\text{new}} = a_{\text{old}} - lr \times g_a$$

$$b_{\text{new}} = b_{\text{old}} - lr \times g_b$$

$$c_{\text{new}} = c_{\text{old}} - lr \times g_c$$